NAMD Installation Guide (Ubuntu and macOS)

CIS-1 tutorial prep · Thursday, 12 Nov 2026, 2:00–3:00 PM, A3 301

Please install NAMD on your laptop **before** the tutorial and confirm it runs using the check at the end of your section.

Before you start (both systems)

1. Create a free account at the NAMD download page: https://www.ks.uiuc.edu/Development/Download/download.cgi?PackageName=NAMD (registration is required to download).
2. Download the **prebuilt binary** (no compiling needed). Use the latest stable **NAMD 3.0.x** release, **multicore** build (runs on a single laptop, no MPI needed).
3. Keep at least a few GB of free disk space and a stable internet connection.
4. Binary names: NAMD 3.x provides `namd3`; NAMD 2.14 provides `namd2`. Use whichever matches your download.

Part A: Ubuntu (20.04 / 22.04 / 24.04, x86_64)

Option 1 (recommended): official prebuilt binary

1. Download the file named like `NAMD_3.0.x_Linux-x86_64-multicore.tar.gz` from the link above.
2. Open a terminal and extract it:

```
cd ~/Downloads
tar -xzf NAMD_3.0.x_Linux-x86_64-multicore.tar.gz
mv NAMD_3.0.x_Linux-x86_64-multicore ~/namd
```

3. Add NAMD to your PATH so you can run it from anywhere:

```
echo 'export PATH=$HOME/namd:$PATH' >> ~/.bashrc
source ~/.bashrc
```

4. Verify:

```
namd3 --help
```

If you see the usage message, the install works. (For NAMD 2.14, run `namd2 --help`.)

Option 2: Ubuntu package (quick, but may be an older version)

```
sudo apt update
sudo apt install namd
namd2 --help
```

This installs whatever version Ubuntu ships, which is typically an older NAMD 2.x. It is fine for basic use, but Option 1 is preferred.

Common problems on Ubuntu

- **Permission denied when running** `namd3`: run `chmod +x ~/namd/namd3 ~/namd/charmrun`.
- `command not found`: PATH not updated. Open a new terminal or run `source ~/.bashrc`, and check `ls ~/namd` shows the `namd3` file.
- **Missing library errors**: run `sudo apt install libgomp1` and try again.
- **Using WSL or a VM**: follow the same steps inside the Ubuntu environment.

Part B: macOS

Steps

1 Check your chip: Apple menu → **About This Mac**. It says either **Apple (M1/M2/M3…)** or **Intel**.

2 From the download page, get the **macOS multicore** build that matches your chip. If only an Intel (x86_64) build is offered for your version, it can run on Apple Silicon through Rosetta 2. Install Rosetta once with:

```
softwareupdate --install-rosetta --agree-to-license
```

3 Extract the downloaded archive (double-click it, or in Terminal):

```
cd ~/Downloads
tar -xzf NAMD_3.0.x_MacOS-*-multicore.tar.gz
mv NAMD_3.0.x_MacOS-*-multicore ~/namd
```

4 macOS may block downloaded programs. Remove the quarantine flag:

```
xattr -dr com.apple.quarantine ~/namd
```

5 Add NAMD to your PATH (macOS uses zsh by default):

```
echo 'export PATH=$HOME/namd:$PATH' >> ~/.zshrc
source ~/.zshrc
```

6 Verify:

```
namd3 --help
```

(For NAMD 2.14, run `namd2 --help`.) If you see the usage message, the install works.

Common problems on macOS

- **"cannot be opened because the developer cannot be verified"**: run the `xattr` command in step 4 again, or go to System Settings → Privacy & Security and click **Allow Anyway**.
- **"bad CPU type in executable"**: you downloaded a build for the wrong chip, or Rosetta is not installed. Install Rosetta (step 2) or download the matching build.
- `command not found`: open a new Terminal window or run `source ~/.zshrc`, and check that `ls ~/namd` shows the `namd3` file.

Final check (before the tutorial)

- `namd3 --help` (or `namd2 --help`) prints the usage message
- Laptop is charged and you have brought the charger
- Optional: install VMD (https://www.ks.uiuc.edu/Research/vmd/) if you want to visualize structures